

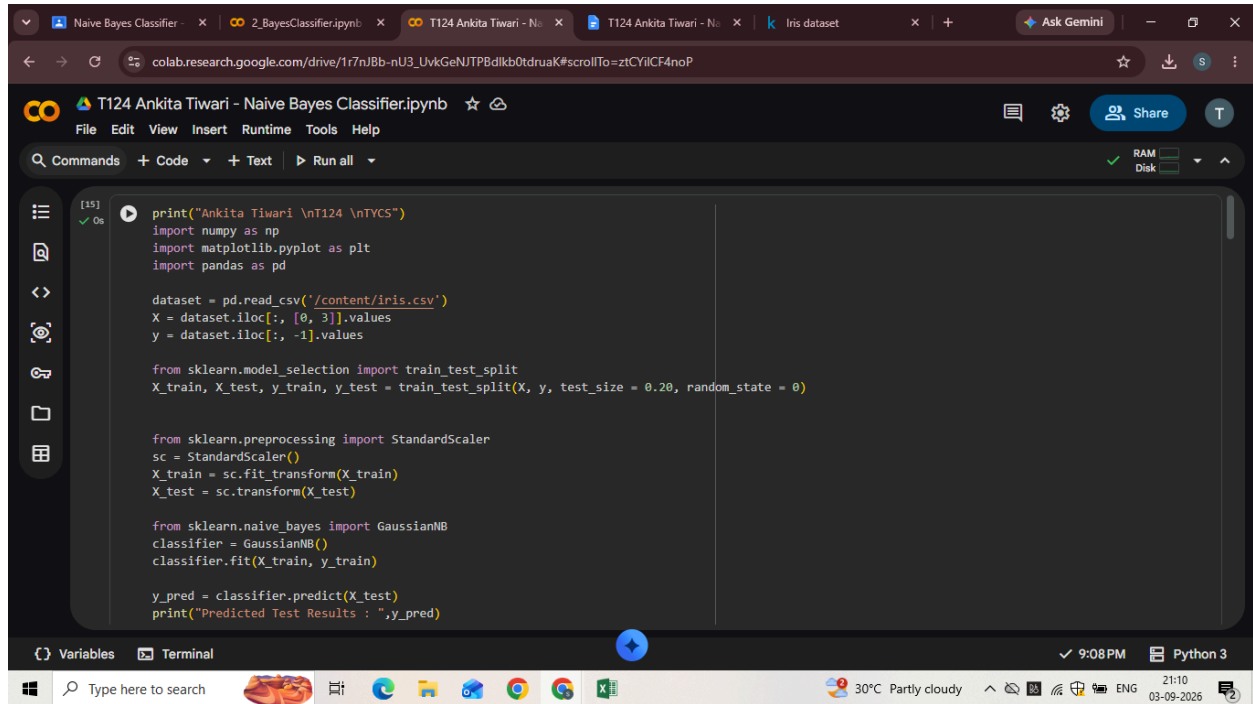
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AIM:

Naive Bayes' Classifier

- Implement the Naive Bayes' algorithm for classification.
- Train a Naive Bayes' model using a given dataset and calculate class probabilities.
- Evaluate the accuracy of the model on test data and analyze the results.

INPUT:



```
[15] print("Ankita Tiwari \nT124 \nTYCS")
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

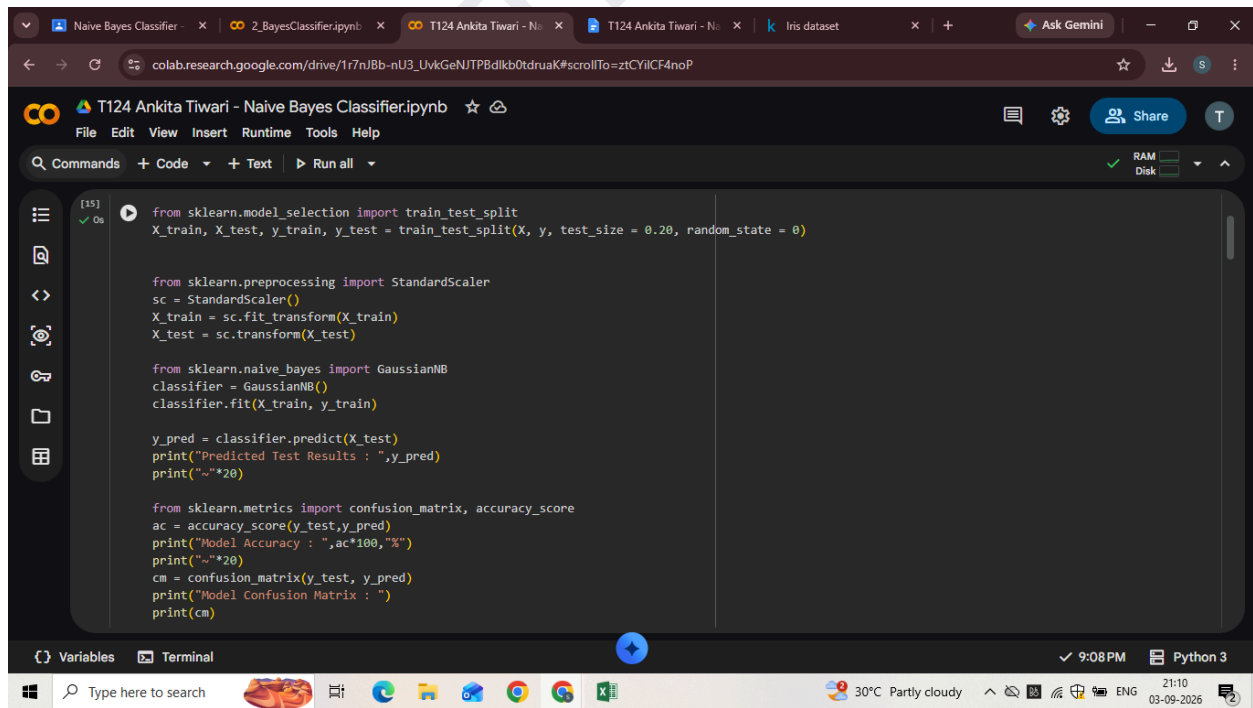
dataset = pd.read_csv("/content/iris.csv")
X = dataset.iloc[:, [0, 3]].values
y = dataset.iloc[:, -1].values

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.20, random_state = 0)

from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)

y_pred = classifier.predict(X_test)
print("Predicted Test Results : ",y_pred)
```



```
[15] from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.20, random_state = 0)

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from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)

y_pred = classifier.predict(X_test)
print("Predicted Test Results : ",y_pred)
print("\n"*20)

from sklearn.metrics import confusion_matrix, accuracy_score
ac = accuracy_score(y_test,y_pred)
print("Model Accuracy : ",ac*100,"%")
print("\n"*20)
cm = confusion_matrix(y_test, y_pred)
print("Model Confusion Matrix : ")
print(cm)
```

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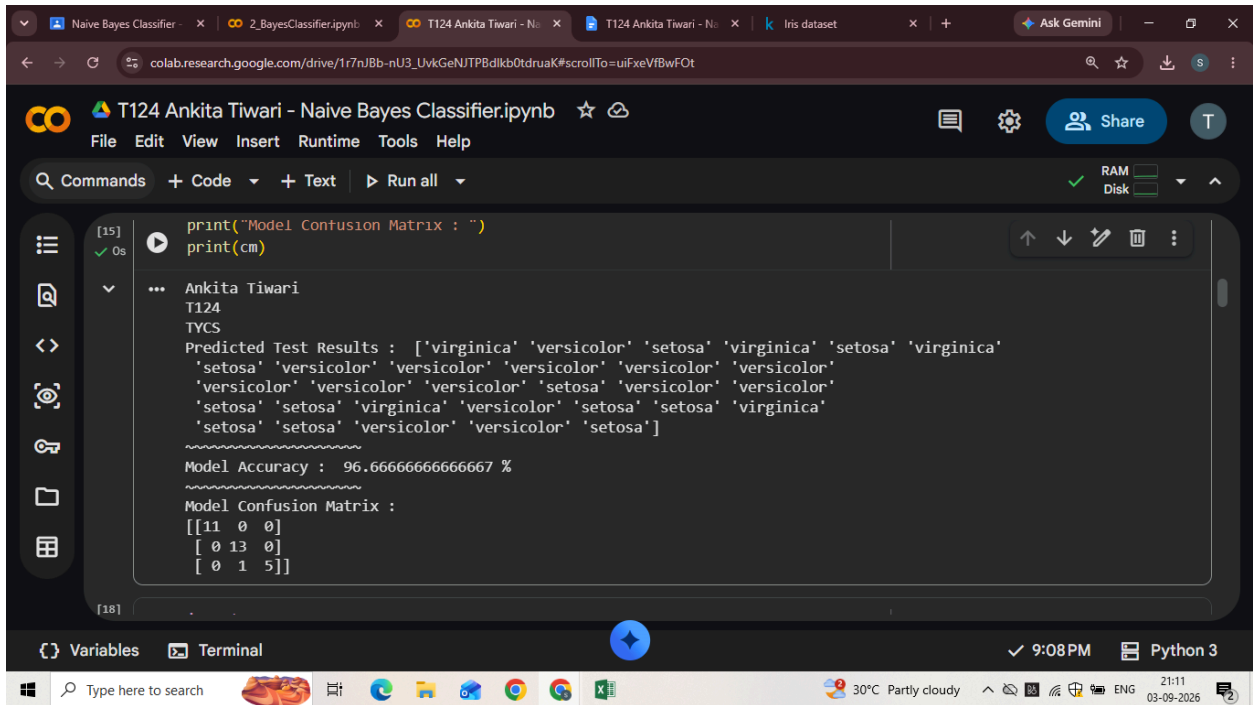
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OUTPUT:

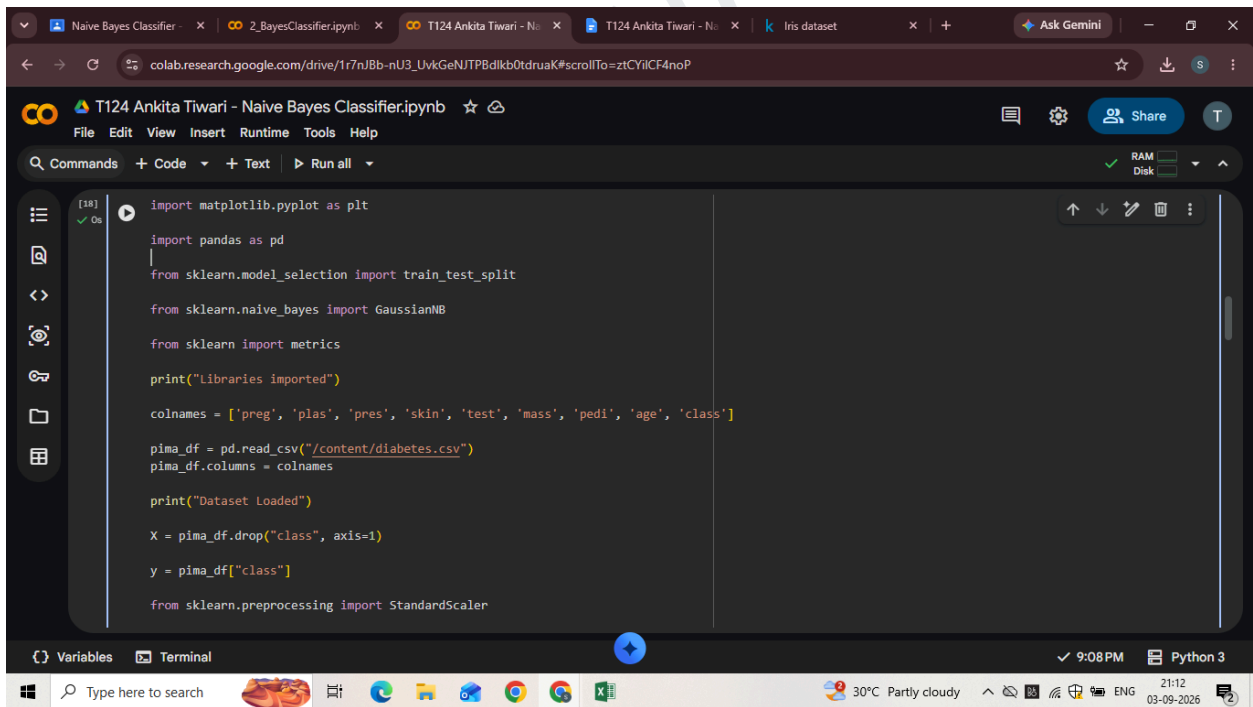


```
[15] print("Model Confusion Matrix : ")
print(cm)

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TYCS
Predicted Test Results : ['virginica' 'versicolor' 'setosa' 'virginica' 'setosa' 'virginica'
'setosa' 'versicolor' 'versicolor' 'versicolor' 'versicolor' 'versicolor'
'versicolor' 'versicolor' 'versicolor' 'setosa' 'versicolor' 'versicolor'
'setosa' 'setosa' 'virginica' 'versicolor' 'setosa' 'setosa' 'virginica'
'setosa' 'setosa' 'versicolor' 'versicolor' 'setosa']

~~~~~
Model Accuracy : 96.66666666666667 %
~~~~~
Model Confusion Matrix :
[[11  0  0]
 [ 0 13  0]
 [ 0  1  5]]
```

INPUT:



```
[18] import matplotlib.pyplot as plt

import pandas as pd
|
from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn import metrics

print("Libraries imported")

colnames = ['preg', 'plas', 'pres', 'skin', 'test', 'mass', 'pedi', 'age', 'class']

pima_df = pd.read_csv("/content/diabetes.csv")
pima_df.columns = colnames

print("Dataset Loaded")

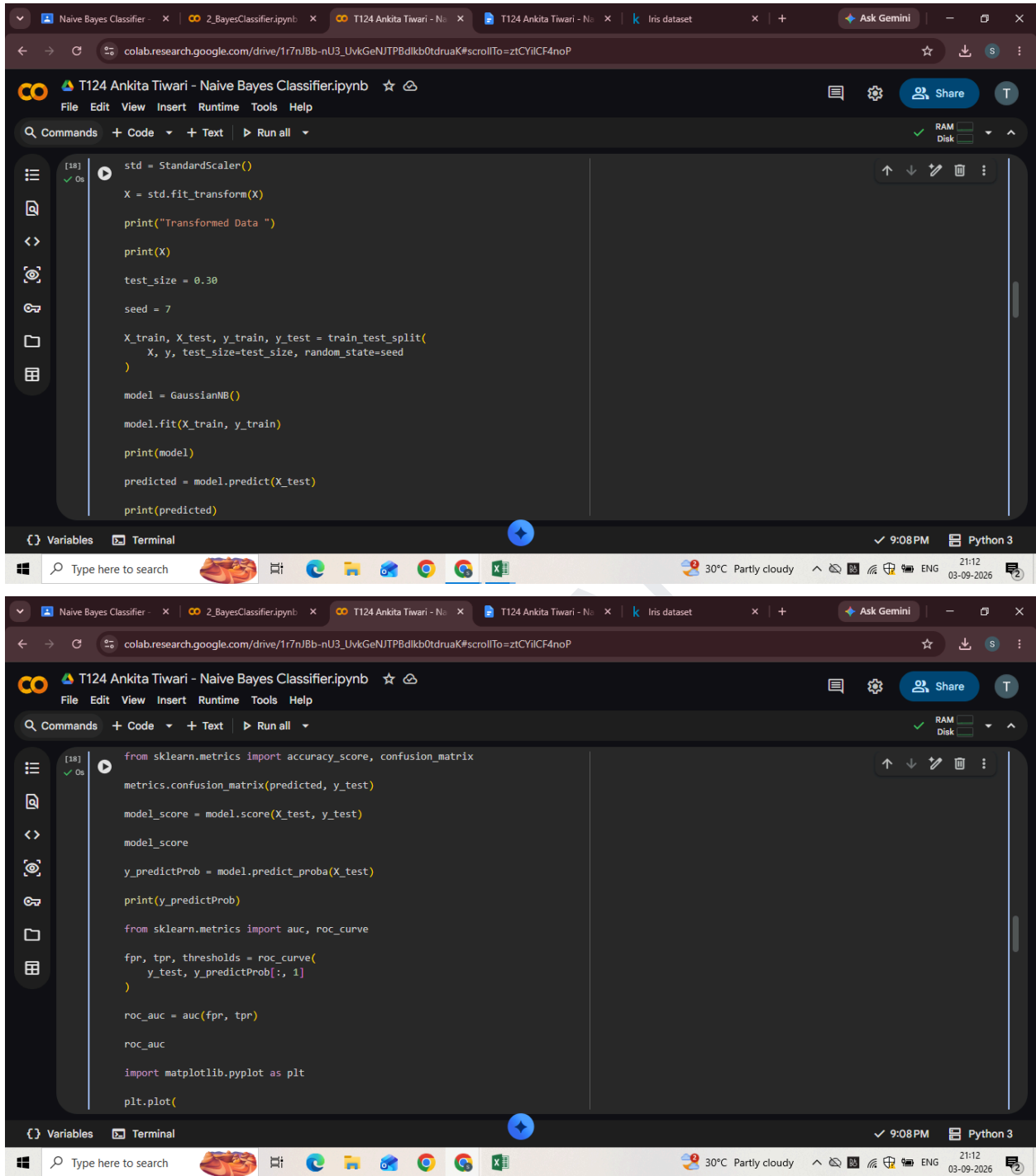
X = pima_df.drop("class", axis=1)

y = pima_df["class"]

from sklearn.preprocessing import StandardScaler
```

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The image displays two screenshots of a Google Colab notebook titled "T124 Ankita Tiwari - Naive Bayes Classifier.ipynb". The notebook is running on a Python 3 environment.

First Screenshot (Top): This screenshot shows the initial steps of the Naive Bayes Classifier implementation. The code includes:

```
[18] std = StandardScaler()
X = std.fit_transform(X)
print("Transformed Data ")
print(X)

test_size = 0.30

seed = 7

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=test_size, random_state=seed
)

model = GaussianNB()
model.fit(X_train, y_train)
print(model)

predicted = model.predict(X_test)
print(predicted)
```

Second Screenshot (Bottom): This screenshot shows the evaluation steps of the classifier. The code includes:

```
[18] from sklearn.metrics import accuracy_score, confusion_matrix

metrics.confusion_matrix(predicted, y_test)

model_score = model.score(X_test, y_test)

model_score

y_predictProb = model.predict_proba(X_test)
print(y_predictProb)

from sklearn.metrics import auc, roc_curve

fpr, tpr, thresholds = roc_curve(
    y_test, y_predictProb[:, 1]
)

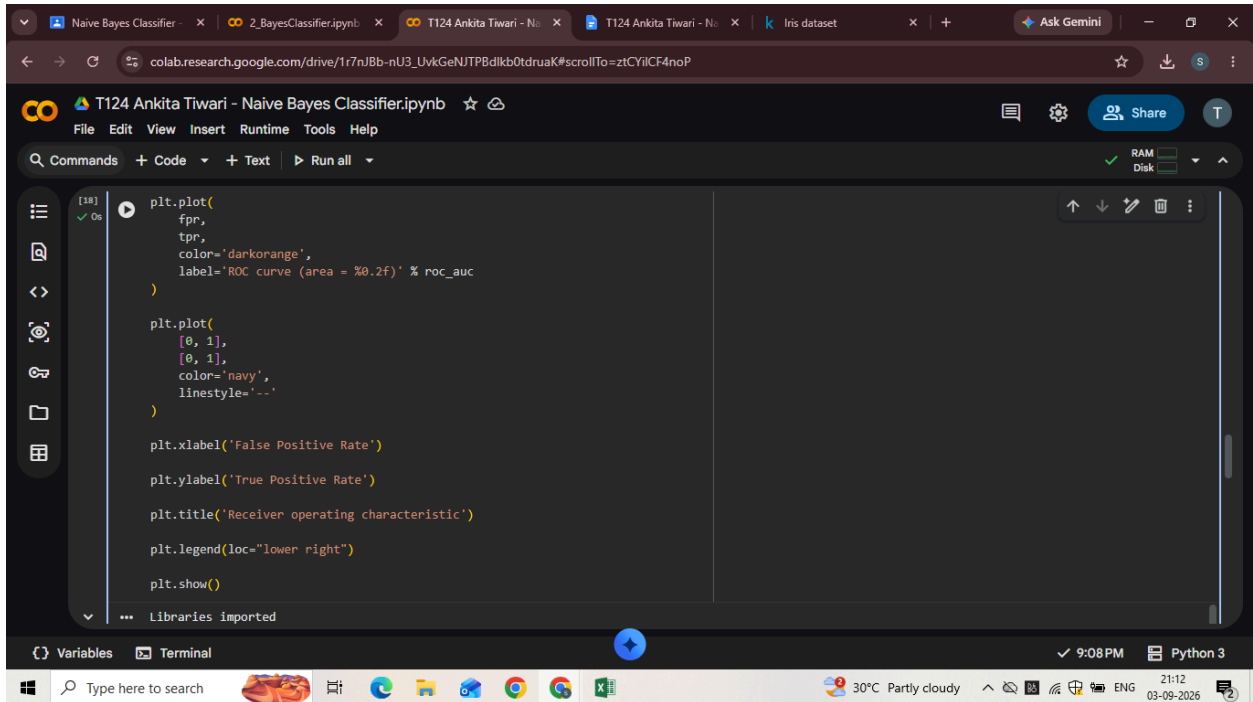
roc_auc = auc(fpr, tpr)

roc_auc

import matplotlib.pyplot as plt
plt.plot(
```

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The screenshot shows a Google Colab notebook titled "T124 Ankita Tiwari - Naive Bayes Classifier.ipynb". The code in the cell is as follows:

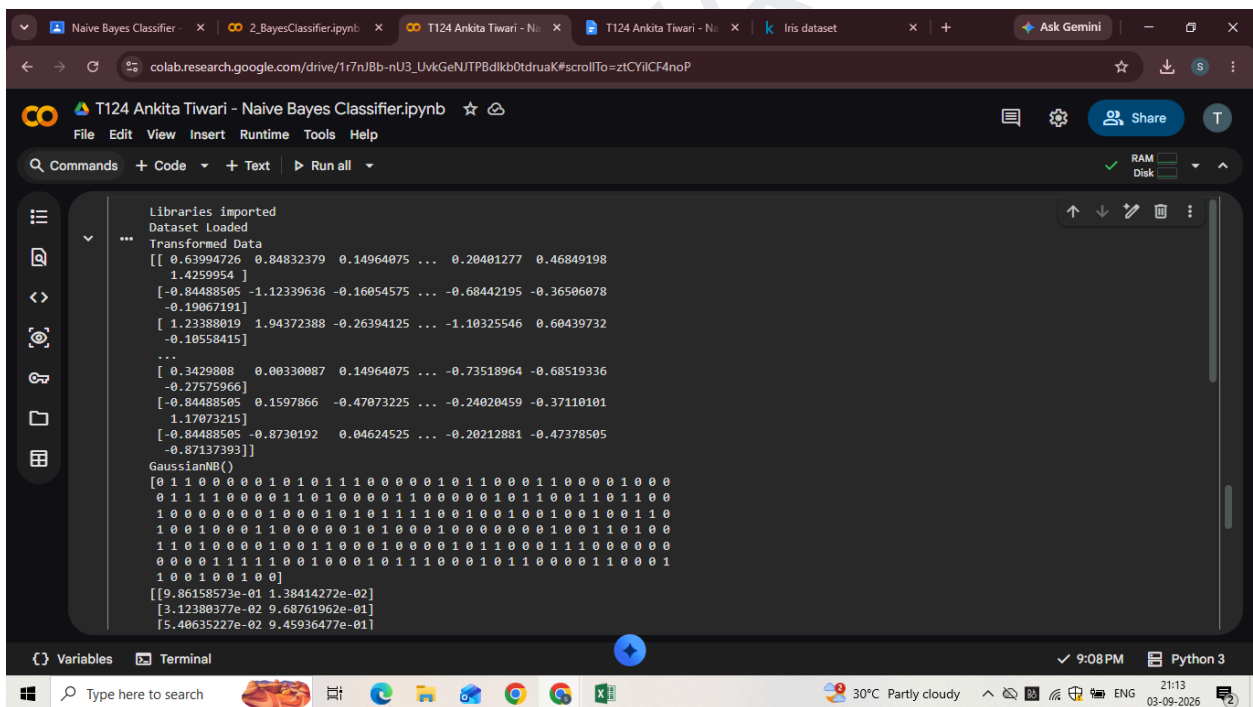
```
[18]: plt.plot(
      fpr,
      tpr,
      color='darkorange',
      label='ROC curve (area = %0.2f)' % roc_auc
    )

    plt.plot(
      [0, 1],
      [0, 1],
      color='navy',
      linestyle='--'
    )

    plt.xlabel('False Positive Rate')
    plt.ylabel('True Positive Rate')
    plt.title('Receiver operating characteristic')
    plt.legend(loc="lower right")
    plt.show()
```

The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations and execution, and a status bar at the bottom showing the time as 9:08 PM and the environment as Python 3.

OUTPUT:



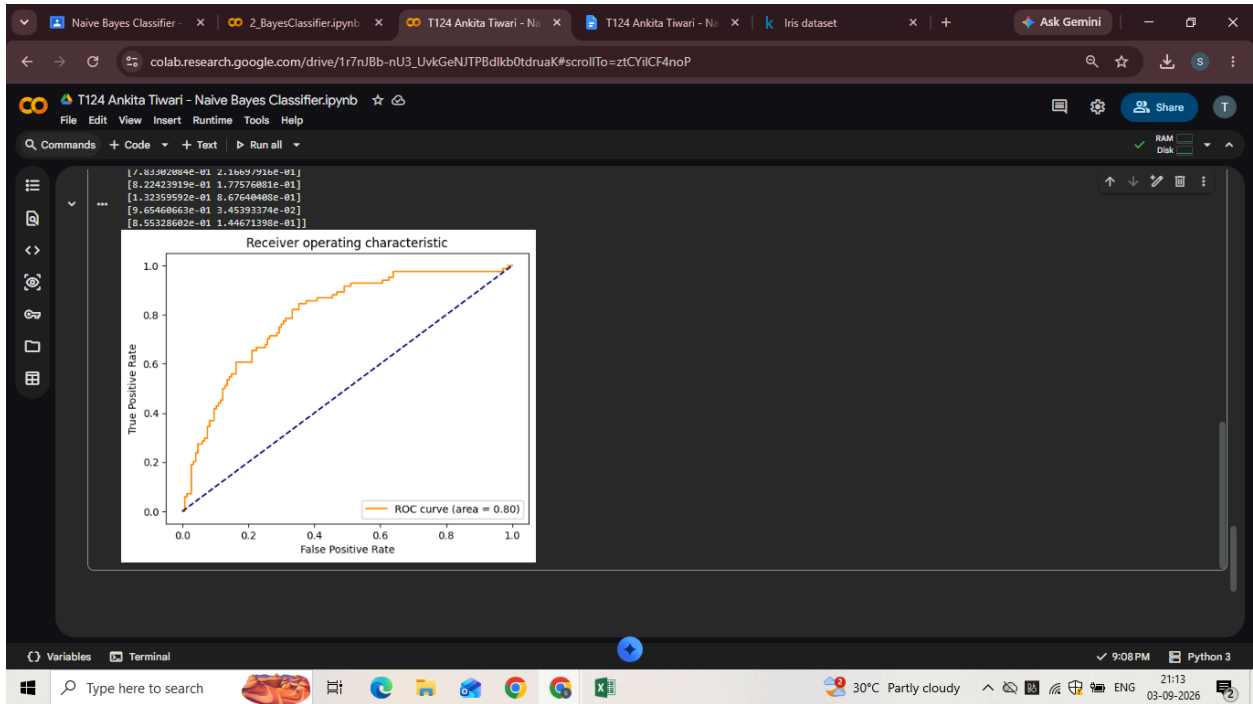
The screenshot shows the same Google Colab notebook, but the output of the code is displayed. The output includes the following text:

```
Libraries imported
Dataset Loaded
Transformed Data
[[ 0.63994726  0.84832379  0.14964075 ... 0.28401277  0.46849198
  1.4259954 ]
 [-0.84488505 -1.12339636 -0.16054575 ... -0.68442195 -0.36506078
 -0.19067191]
 [ 1.23388019  1.94372388 -0.26394125 ... -1.10325546  0.60439732
 -0.10558415]
 ...
 [ 0.3429888  0.00330087  0.14964075 ... -0.73518964 -0.68519336
 -0.27575966]
 [-0.04488505  0.1597866  -0.47073225 ... -0.24020459 -0.37110101
  1.17073215]
 [-0.04488505 -0.8730192  0.04624525 ... -0.20212881 -0.47378505
 -0.87137393]]
GaussianNB()
[[0 1 1 0 0 0 0 1 0 1 0 1 1 1 0 0 0 0 0 1 0 1 1 1 0 0 0 0 1 1 0 0 0 0 1 0 0 0
 0 1 1 1 1 0 0 0 0 1 1 0 1 0 0 0 0 1 1 0 0 0 0 0 1 0 1 1 1 0 0 1 1 0 1 1 0 0
 1 0 0 0 0 0 0 1 0 0 0 1 0 1 0 1 1 1 1 0 0 1 0 0 1 0 0 1 0 0 1 0 0 1 1 0
 1 0 0 1 0 0 0 1 1 0 0 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 0 0 1 0 0 1 1 0 1 0
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 0 0 0 0 1 1 1 1 1 0 0 1 0 0 0 1 0 1 1 1 0 0 0 1 0 1 1 0 0 0 0 1 1 0 0 0
 1 0 0 1 0 0 1 0 0]
[[9.86158573e-01 1.38414272e-02]
 [3.12380377e-02 9.68761962e-01]
 [5.40635227e-02 9.45936477e-01]]
```

The interface is identical to the first screenshot, showing the same menu bar, toolbar, and status bar.

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